

Library Project: SQL Query Handbook

Implementation scripts for tracking, insights, and recommendations.

Section 1: Data Definition (DDL)

Creating the foundational structures for our library system.

```
-- Create table for inventory
CREATE TABLE Books (
  book_id INT PRIMARY KEY AUTO_INCREMENT,
  title VARCHAR(150) NOT NULL,
  author_id INT,
  category_id INT,
  available_copies INT DEFAULT 5,
  FOREIGN KEY (author_id) REFERENCES Authors(author_id)
);
```

Section 2: Library Insights & Analytics

Queries to help librarians understand borrowing trends.

```
-- 1. Identify Most Borrowed Books
SELECT b.title, COUNT(br.borrow_id) AS total_borrows
FROM Books b
JOIN Borrow_Records br ON b.book_id = br.book_id
GROUP BY b.book_id
ORDER BY total_borrows DESC
LIMIT 10;

-- 2. Find Members with Overdue Books
SELECT m.member_name, b.title, br.due_date
FROM Borrow_Records br
JOIN Members m ON br.member_id = m.member_id
JOIN Books b ON br.book_id = b.book_id
WHERE br.return_date IS NULL
AND br.due_date < CURDATE();
```

Section 3: Recommendation Engine Logic

Advanced SQL logic to personalize the user experience.

```
-- Suggest books based on a user's most read category
WITH UserFavCategory AS (
  SELECT b.category_id
  FROM Borrow_Records br
  JOIN Books b ON br.book_id = b.book_id
  WHERE br.member_id = :target_user_id
```

```
        GROUP BY b.category_id
        ORDER BY COUNT(*) DESC
        LIMIT 1
    )
    SELECT title, available_copies
    FROM Books
    WHERE category_id = (SELECT category_id FROM UserFavCategory)
    AND book_id NOT IN (
        SELECT book_id FROM Borrow_Records WHERE member_id = :target_user_id
    );
```

Section 4: Automated Business Logic

Using triggers to keep inventory updated automatically.

```
-- Automatically decrease stock when a book is borrowed
CREATE TRIGGER trg_after_borrow
AFTER INSERT ON Borrow_Records
FOR EACH ROW
BEGIN
    UPDATE Books
    SET available_copies = available_copies - 1
    WHERE book_id = NEW.book_id;
END;
```